Team Name: Response 200

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Problem Statement: Intelligent Candidate Discovery & Ranking Challenge - Designing a high-performance candidate shortlisting pipeline that excludes low-quality/trap profiles, scores core ML/Search skills, and aligns candidates by product fit and platform availability.

Solution Overview

Proposed Solution Overview

A high-performance, 100% offline, deterministic rule-based ranking engine designed to source top Senior AI Engineers.
Incorporates deep semantic skill alignment, chronological career parsing, and multi-factor behavioral signal weighting.
Operates on streaming principles to process large datasets (100,000 candidates) in seconds with extremely low memory requirements.

Differentiators from Traditional Systems

Active Fraud & Honeypot Protection: Programmatically flags and excludes candidates with anomalous or logically impossible profile details (e.g. expert skill with 0 duration, startup dates prior to launch).
Availability-Centric Ranking: Factors in recruiter response rate, avg response time, notice period, and active recency. A technically perfect candidate who is inactive gets down-weighted, prioritizing available and active candidates.
Target ML Experience Calculation: Computes actual ML/AI tenure from job history details, avoiding keyword-stuffers and non-technical profiles.

JD Understanding & Candidate Evaluation

- ## Key Requirements Extracted from Job Description (JD)

Experience: 5–9 years (ideally 6–8 years) with a strong preference for product engineering experience.
Technical Core: Vector search, embeddings, vector databases (Pinecone, Milvus, Weaviate), learning-to-rank, evaluation metrics (NDCG, MRR, MAP), and NLP/Deep Learning.
Logistics: Location in or willingness to relocate to Pune/Noida offices (hybrid), and notice periods of sub-30 days preferred.

Evaluating Candidate Fit Beyond Keyword Matching

Chronological Verification: Verifies candidate career history to calculate the actual duration of ML/AI experience rather than relying on raw years of experience.
Title & Role Relevance: Backend/SWE profiles are graded based on their actual ML tenure, and non-technical titles (marketing, HR, operations) are filtered out.
Logical Consistency: Automatically discards candidate profiles containing expert/advanced skills with 0 duration or employment at newly founded startups prior to their actual launch date (honeypots).

Ranking Methodology

- Candidate Retrieval and Filtering

Stream-loads the candidate JSONL pool and immediately filters out (1) Honeypots and (2) Consulting-only backgrounds (candidates who worked ONLY at TCS, Infosys, Wipro, Accenture, Cognizant, etc.).

Multi-Factor Profile Fit Scoring (Weighted 60% of Total)

Experience Fit (15%): Optimal score for 5–9 years of experience, peaking at 6–8 years.
ML Experience (25%): Scoring based on years specifically spent in ML, AI, Search, Retrieval, or Ranking roles.
Role Relevance (20%): Current title graded (AI/ML Eng = 1.0, Data Scientist = 0.9, Backend/SWE = 0.8/0.4, Non-tech = 0.0).
Skill Match (20%): Computes a weighted score based on core search/retrieval/ML skills, adjusted by proficiency level and duration.
Keywords (10%): Scans descriptions for specific technical terms (ndcg, Pinecone, RAG, etc.).
Product Company Ratio (10%): Proportion of career spent at recognized product companies.

Behavioral Signals Calibration (Weighted Multipliers & Bonuses)

Recency & Activity: Multiplies profile fit based on activity recency (up to 0.4x penalty for >6 months inactive) and response rates.
Notice Period: Deducts up to 0.5x for notice periods exceeding 60-90 days, while awarding bonuses (+3% score) for

Explainability & Data Validation

- ## Explainability Engine

Generates structured, 2-3 sentence rationales for every candidate in the shortlist based on their rank band. Mentions total experience, ML experience, specific top core skills, previous product companies, location status, and notice period.
Example: "Exceptional founding engineer candidate with 7.2 years experience, specializing in skills in Deep Learning, Weaviate, Recommendation Systems. Proven record building ranking/retrieval systems at Zomato, Google. locally based in Noida with a sub-30 day notice (15 days)."

Preventing Hallucinations

100% Deterministic NLG: Replaced probabilistic generative models with a rule-based Natural Language Generation (NLG) engine. Every claim in the rationale is directly bound to raw database values, ensuring zero hallucinations and perfect auditability.

Suspicious Profiles & Fraud Handling

Honeypots Rejection: Identifies profiles containing impossible durations (0 months for expert skills) or impossible timelines (Krutrim/Sarvam employment before 2023).
Service/Consulting Rejection: Disqualifies candidates whose entire career history is limited to services companies (TCS, Infosys, Accenture, etc.), keeping only candidates with product experience

End-to-End Workflow

- End-to-End Execution Flow

1. Initialization: Load JD configurations, target skills, consulting blacklist, product company list, and target cities.
2. Streaming Ingestion: Reads the 100,000 candidate dataset directly from candidates.jsonl.gz line by line to keep RAM usage under 100MB.
3. Hard Filtering: Evaluates honeypots and consulting-only profiles. Disqualifies invalid candidates immediately.
4. Feature Extraction: Computes overall experience, ML experience, role title fit, skill score, and product company ratio.
5. Behavioral Scoring: Applies bonuses for active engagement and multipliers (penalties) for long notice periods or lack of relocation.
6. Stable Sorting: Sorts candidates by final score (descending) with candidate_id (ascending) as a tie-breaker to ensure stable ordering.
7. Rationale Generation & Export: Runs the NLG engine on the top 100 candidates to create personalized explanations. Writes the output to submission.csv.

System Architecture

Modular Pipeline Architecture

Data Layer: Handles streaming ingestion of compressed candidate records (gzipped JSONL) to maintain a sub-100MB RAM footprint.
Quality Guardrails (Hard Rejection Filters):
Honeypot/Logic-Trap Detector: Rejects profile inconsistencies (e.g. Krutrim before 2023, expert skill with 0 duration).
Services Company Exclusion: Rejects candidate profiles containing only services/consulting firms.
Multi-Factor Scoring Core:
Technical Fit: Grades overall experience, ML experience, role title fit, skill score, and description keywords.
Product Fit: Computes career ratio at recognized product companies vs. service firms.
Behavioral Calibration (Availability & Location Multipliers):
Computes multipliers and bonuses for activity recency, response rates, response time, relocation interest, and notice period.
Sorter & NLG Output Engine:
Stable Sorter: Sorts by score descending, with candidate ID as secondary tiebreaker.
NLG Template Generator: Forms structured, fact-based candidate justifications.
Exporter: Outputs the top 100 candidate shortlist to submission.csv.

Results & Performance

Key Quantitative Results & Shortlist Quality

Ingestion & Filtering: Ingested 100,000 candidate profiles. Hard filters rejected 9.84% of candidates, leaving 90,159 valid profiles for final ranking.
Top shortlists showcase exceptional candidates fitting all JD requirements:
Aarav Trivedi (#1, Score: 1.1600): 7.2 yrs exp, 7.2 yrs ML exp, ex-Zomato. Skills: Deep Learning, Weaviate, Recommendation Systems. Locally based in Noida, sub-30 day notice (15 days).
Nisha Pillai (#2, Score: 1.1500): 7.6 yrs exp, 7.6 yrs ML exp, ex-Sarvam AI. Skills: scikit-learn, PyTorch, Milvus. Locally based in Gurgaon, 45-day notice.
Ayaan Goyal (#3, Score: 1.1400): 6.5 yrs exp, 6.3 yrs ML exp, ex-Haptik. Skills: LoRA, Recommendation Systems, Weaviate. Locally based in Pune, sub-30 day notice (15 days).

Compute & Resource Performance

Runtime Efficiency: Processed and ranked all 100,000 candidates in ~12.5 seconds (against the 5-minute challenge limit, a 24x speedup).
Memory footprint: Streaming reader keeps RAM usage under 100MB (against the 16GB limit, a 160x memory reduction).
Fully Offline: 100% local CPU processing, requiring no network calls or external API keys, meeting all offline

Technologies Used

Development Technologies & Frameworks

Python 3.12: The primary programming language, selected for its speed in scripting, robust text parsing, and standard library data tools.
NumPy: Leveraged for high-performance numerical operations and precision rounding of candidate scores.
gzip & json: Standard streaming libraries used to parse candidates.jsonl.gz directly, preventing disk overhead and keeping RAM footprint minimal.
datetime: Used for chronological checks (e.g. startup inception dates, activity recency decay).
csv: Utilized to write standard-compliant, RFC-conformant output files.

AI & Automation Tools

Gemini 3.5 Flash via Antigravity: Used for parsing raw schemas, design review of honeypot filters, validating requirements, and ensuring 100% compliance with output specs.
python-pptx: Programmatic library utilized to insert text and structure slides within the presentation template.

Submission Assets

Primary Submission Deliverables

GitHub Repository: https://github.com/vishwabhishek/Resume_run
Live Interactive Sandbox: Live environment running on Hugging Face Spaces at: https://huggingface.co/spaces/abhishek-vishwakarma/redrob-ranker
Pipeline Entry Point: rank.py (The execution script that processes scoring, filtering, and exports results in seconds).

Generated Artifacts

submission.csv: Contains the final ranked top 100 candidates with IDs, scores, and natural language explanations.
submission_metadata.yaml: Declares team metadata, parameters, compute environments, and reproduction instructions.

INDIA.RUNS

Build what next India runs on

THANK YOU